TOPS Technologies

Interview Questions — Django REST Framework (REST API Development)

Total Questions: 47

---

1. What is an API and why are APIs important in web development?

> **Hint:** Think about how different systems communicate with each other.

---

2. Explain what a REST API is and name two key principles it follows.

> **Hint:** Consider statelessness and how resources are handled.

---

3. What is the difference between JSON and XML when it comes to data exchange in APIs?

> **Hint:** Focus on readability and typical use in modern APIs.

---

4. How is Django REST Framework (DRF) different from standard Django?

> **Hint:** Think about what DRF adds to Django for API development.

---

5. Describe the steps to install Django REST Framework in a Django project.

> **Hint:** Consider both package installation and settings configuration.

---

6. What is the purpose of adding 'rest_framework' to INSTALLED_APPS in Django settings?

---

7. What is the difference between a Serializer and a ModelSerializer in DRF?

> **Hint:** Think about which one is more automatic and when you’d use each.

---

8. How would you create a simple 'Hello API' endpoint in DRF using APIView?

---

9. What is a Django model, and why do we define models before creating API endpoints?

> **Hint:** Consider how models relate to the database.

---

10. Write a basic DRF Serializer for a model called Doctor with fields 'name' and 'specialization'.

> **Hint:** Remember to import Serializer and define fields.

---

11. How do you create a CRUD API endpoint for a model using APIView in DRF?

> **Hint:** Think about the HTTP methods involved and their mapping.

---

12. What is the difference between HTTP status codes 200 and 201 in API responses?

> **Hint:** One is for successful retrieval, the other for successful creation.

---

13. How would you return a JSON response from a DRF API endpoint?

---

14. Write the code for a POST endpoint to create a new Doctor using ModelSerializer in DRF.

---

15. What is the use of GenericAPIView and Mixins in DRF?

> **Hint:** Think about reducing code repetition for common operations.

---

16. Describe how you would update a Doctor object using a PUT request in DRF.

---

17. Explain what a ViewSet is in DRF and how it differs from APIView.

> **Hint:** Focus on how ViewSets group logic and reduce code.

---

18. What is the purpose of a Router in DRF, and what does it generate automatically?

> **Hint:** Think about URL patterns for your API.

---

19. Explain the difference between DefaultRouter and SimpleRouter in DRF.

---

20. How does pagination help when returning large querysets from an API?

> **Hint:** Consider both server performance and client usability.

---

21. Describe how you would implement PageNumberPagination for a list endpoint in DRF.

---

22. What is the difference between LimitOffsetPagination and CursorPagination in DRF?

---

23. How would you add ordering by 'name' to a list endpoint in DRF?

---

24. What are the main types of authentication available in DRF?

> **Hint:** List at least three and describe them briefly.

---

25. Explain the difference between TokenAuthentication and SessionAuthentication in DRF.

> **Hint:** Consider where the authentication information is stored.

---

26. Write a DRF permission class that only allows access to authenticated users.

---

27. How would you protect the /api/doctors/ endpoint so that only logged-in users can access it?

---

28. What is the role of the IsAdminUser permission class in DRF?

---

29. If you wanted to create a custom permission that only allows doctors with a certain specialization to edit their profile, how would you approach this in DRF?

> **Hint:** Think about overriding the 'has_permission' or 'has_object_permission' methods.

---

30. How would you generate and use an authentication token for a user in DRF?

---

31. What is JWT and how is it different from basic token authentication?

> **Hint:** Think about the structure and usage of JWTs.

---

32. How could you use ChatGPT or Copilot to help you explain the difference between Token and Session Authentication in an interview? What would you check before trusting the output?

> **Hint:** Think about verifying explanations and cross-checking with official docs.

---

33. Describe the steps to integrate an external API, such as OpenWeatherMap, into a DRF endpoint.

---

34. Write a DRF view that calls an external weather API and returns the temperature for a given city.

---

35. How would you handle errors if the external API you called from your DRF endpoint is down or returns invalid data?

> **Hint:** Consider exception handling and response status codes.

---

36. What is Postman and how can it help you test your DRF API endpoints?

> **Hint:** Focus on features that make API testing easier.

---

37. How could you use Postman AI or ChatGPT to generate code snippets for calling your DRF API from Python?

> **Hint:** Think about the 'Code' feature and checking if the code matches your API's requirements.

---

38. Explain how you would send an email using an external service like Mailgun or MailChimp from a DRF endpoint.

---

39. Describe how you would integrate Twilio to send an SMS notification from your DRF API.

---

40. What steps are involved in deploying a DRF API to PythonAnywhere?

> **Hint:** Think about preparing your project, setting up the web app, and configuring settings.

---

41. Why is API versioning important, and how would you implement it in DRF?

> **Hint:** Consider backward compatibility and URL structure.

---

42. How would you format a custom JSON response in DRF, for example, to wrap all data in a 'result' key?

---

43. Describe how exception handling in DRF can improve the robustness of your API.

---

44. Imagine you have deployed your DRF API and a client reports that the /api/doctors/ endpoint returns a 500 error. How would you troubleshoot this?

> **Hint:** Think about checking logs, error messages, and recent code changes.

---

45. Integrative: How would you secure an endpoint that integrates with an external payment API, ensuring only authenticated users can access it and handling errors gracefully?

> **Hint:** Combine authentication, permissions, and exception handling.

---

46. Integrative: Describe how you would create a paginated, authenticated endpoint that lists doctors and allows filtering by specialization.

---

47. Integrative: Suppose you need to update your API to v2, adding a new required field to the Doctor model. How would you handle this change to avoid breaking existing clients?

> **Hint:** Think about versioning and backward compatibility.
